Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories – spending patterns, segments, preferences, and subscriptions.

Dataset at a Glance

3,900	18	37	$59.76
Purchases	***Columns***	***Missing Values***	***Avg Purchase***
Total transactions analyzed	Features per record	In Review Rating column	Mean spend per transaction

Key features: demographics, purchase details, shipping, discounts, review ratings, and subscription status.

Data Preparation in Python

01

Load and Explore

`pandas`, `df.info()`, `.describe()`

02

Clean and Impute

Median imputation for missing Review Ratings; snake_case columns

03

Feature Engineering

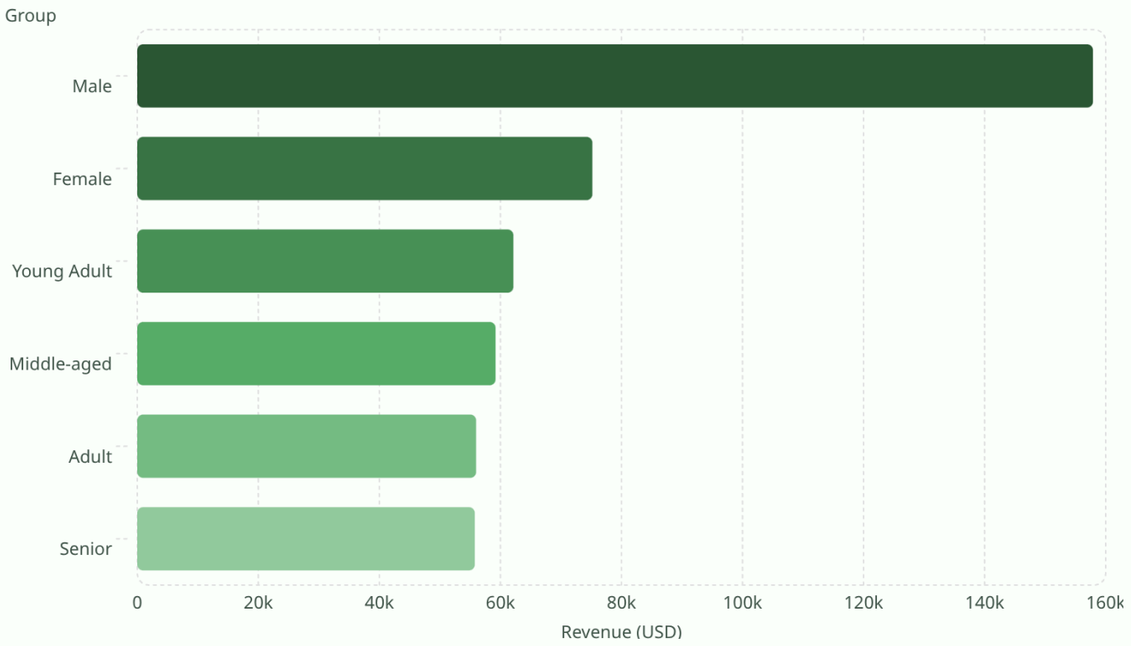
Added `age_group` and `purchase_frequency_days`

04

Load to Microsoft SQL Server

Cleaned DataFrame exported to database for SQL analysis

Revenue and Spending Insights



Key Findings

- Male customers generate **2× more revenue** than female
- Young Adults lead by age group at **$62K**
- Express shipping users avg **$60.48** vs Standard **$58.46**
- 863 discount users still spent **above average**

Top Products and Categories

🏆 Top Rated

- Gloves – 3.86
- Sandals – 3.84
- Boots – 3.82

👗 Clothing

- Blouse – 171
- Pants – 171
- Shirt – 169

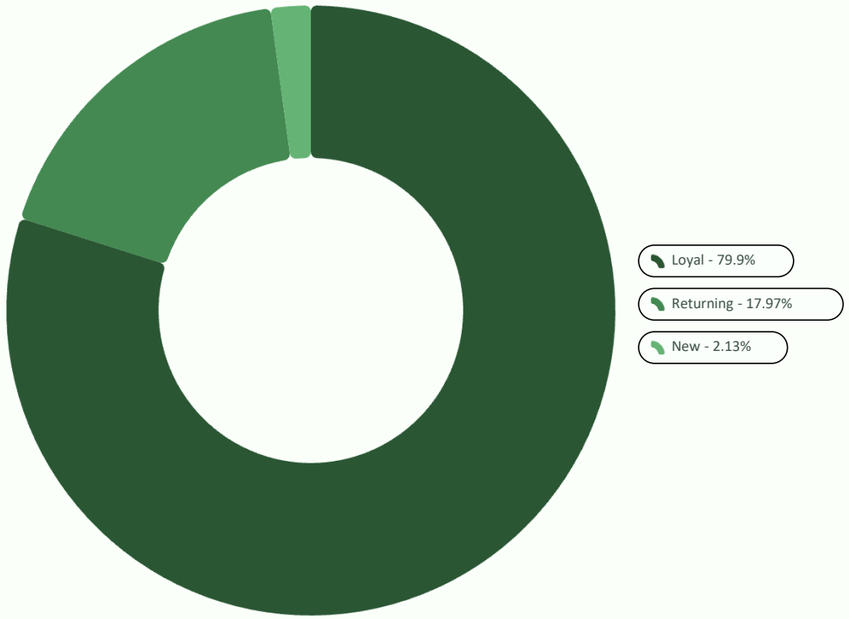
👟 Footwear

- Sandals – 160
- Shoes – 150
- Sneakers – 145

💍 Accessories

- Jewelry – 171
- Belt – 161
- Sunglasses – 161

Customer Segmentation



Segment Breakdown

Based on purchase history – New, Returning, and Loyal.

Loyal

3,116 customers

Returning

701 customers

New

83 customers

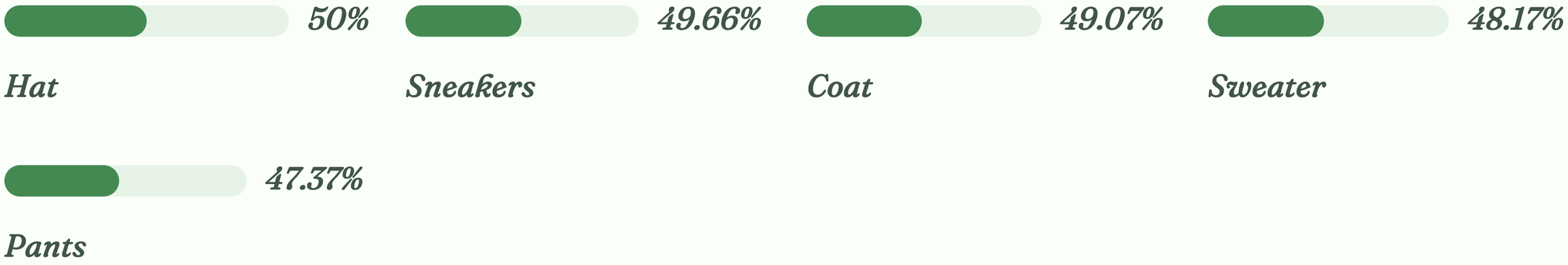
Subscriptions and Discounts

Subscribers vs. Non-Subscribers

Status	Avg Spend	Total Revenue
Subscriber	$59.49	$62,645
Non-Subscriber	$59.87	$170,436

Repeat buyers (>5 purchases): **958 subscribers** vs. 2,518 non-subscribers.

Most Discount-Dependent Products



Power BI Dashboard

3.9K

Customers

$59.76

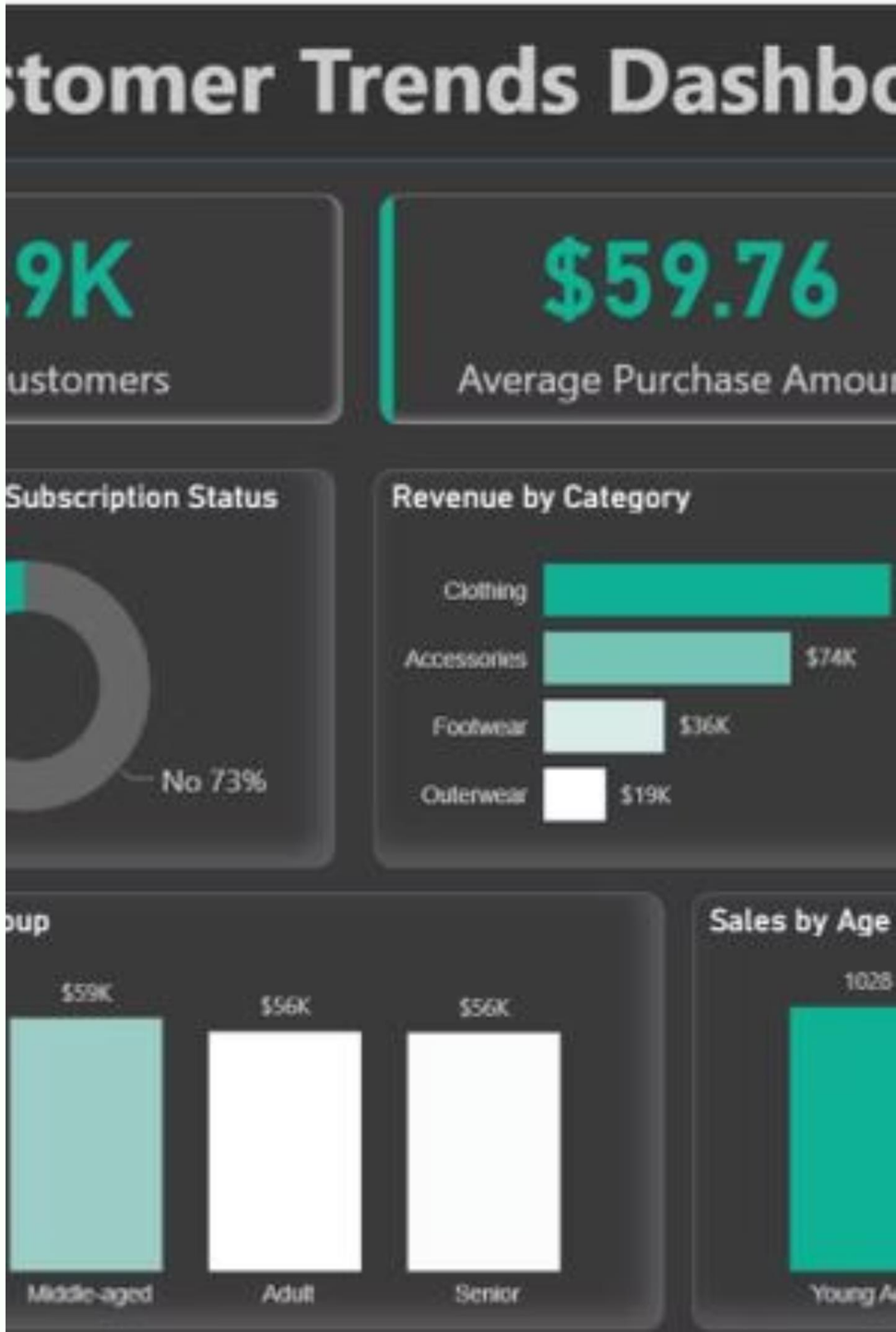
Avg Purchase

3.75

Avg Rating

73% No / 27% Yes

Subscription Split



Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated items in campaigns

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

Key Takeaways

Data-Driven Foundation

3,900 transactions cleaned, engineered, and loaded to Microsoft SQL Server via Python

Dominant Segments

80% of customers are **Loyal**; Young Adults drive the most revenue

Subscription Opportunity

Only 27% subscribe – significant growth potential with targeted incentives

Discount Risk

Several top products rely heavily on discounts – margin review needed